# SMART INDIA HACKATHON 2026

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- **Problem Statement ID –** 26034
- **Problem Statement Title –** Software System to check compliance of packaged commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning product images and labels.
- **Theme –** Miscellaneous
- **PS Category –** Software
- **Team ID –** K26234
- **Team Name –** HexTech
- **Solution –** LMCS · Legal Metrology Compliance Scanner




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# PROPOSED SOLUTION

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## The problem

- Every pack has to show the maker, net quantity, MRP, dates and customer care (Rule 6), and right now all of it is checked by hand.
- A changed MRP or quantity still looks like a normal label, so just reading it isn't enough. We have to check it against the real product details first.

## What makes LMCS different

### We check the label against a registry

- We look up the barcode in our product registry
- Then compare MRP, quantity, FSSAI licence, GSTIN, veg mark and maker
- If anything doesn't match, we flag it as possible tampering

### Evidence and verification

- Each photo gets a SHA-256 fingerprint, GPS location and exact time
- Once a scan is sent, nobody can edit it
- No notice goes out until the inspector and Controller both agree

**Every scan also gets:** all 13 rule checks (unit sale price too), the FSSAI veg mark, print contrast and font size. If a reading is borderline, we mark it REVIEW.

## How it works, step by step

**1. A consumer scans the label** — CONSUMER
The phone reads the label text and barcode by itself

**2. 13 rule checks and a registry match** — AUTOMATIC
Rules 6, 9 and 27, plus the printed MRP, quantity and licence vs. what's registered

**3. Complaint shows up on the inspector's map** — AUTOMATIC
Only when a check fails or details don't match, never when we just can't tell

**4. Inspector checks on site, files a report** — INSPECTOR
The report can only include checks that actually failed

**5. Controller reviews it, confirms or rejects** — CONTROLLER
They see the photo, the fingerprint check, GPS and time

**6. Controller sends the GO notice (PDF)** — CONTROLLER
Only after confirming, and only one notice per report

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# TECHNICAL APPROACH

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## What we used, and why

### On the phone

**Flutter · camera**
One Android app for both consumers and inspectors; it takes the label photo

**Google ML Kit Text Recognition v2**
Reads all the label text right on the phone, and where each line sits

**Google ML Kit Barcode Scanning**
Reads the barcode number and where it is in the photo

**Field extractor (our code)**
Matches each heading to its value by position, even when OCR slips, like “Rs.109,00”

**geolocator · ntp**
Gets the GPS location, and checks the phone's clock against pool.ntp.org

**Dio · secure storage**
Sends the photo and readings to the server, and keeps the login safe

### On the server (Python)

**FastAPI**
Takes in every scan and checks who's asking before doing anything

**OpenCV + NumPy**
Spots the FSSAI veg mark by its shape and colour, and measures print contrast (Otsu, WCAG)

**pyzbar (ZBar) · backup barcode reader**
If the phone can't read the barcode, the server tries again on the photo

**Rules engine (our code)**
Runs 13 checks; each says PASS, FAIL, REVIEW or EXEMPT and names its rule

**Registry cross-check (our code)**
Uses the barcode to pull up the registered MRP, quantity, licence, GSTIN, veg mark and maker

**SHA-256 · JWT · bcrypt**
Fingerprints each photo, hashes passwords, and only an admin can create officer accounts

### Data, maps and documents

**PostgreSQL**
Stores every scan, check, complaint, report and notice; row locks stop double notices

**SQLAlchemy · Alembic**
Our database models, with 8 tracked schema changes

**React + Vite**
The Controller's dashboard: queue, evidence, search, and PDF and Excel downloads

**Leaflet · flutter_map · OpenStreetMap**
Maps of complaints and reports, on the dashboard and on the inspector's phone

**ReportLab**
Makes two PDFs: the compliance report (with the photo) and the GO show-cause notice

**pytest · flutter test**
287 backend and 75 mobile tests that run automatically

**Each check gives** PASS FAIL REVIEW EXEMPT — Only a FAIL can go into a report. REVIEW just means a person should look at the pack.

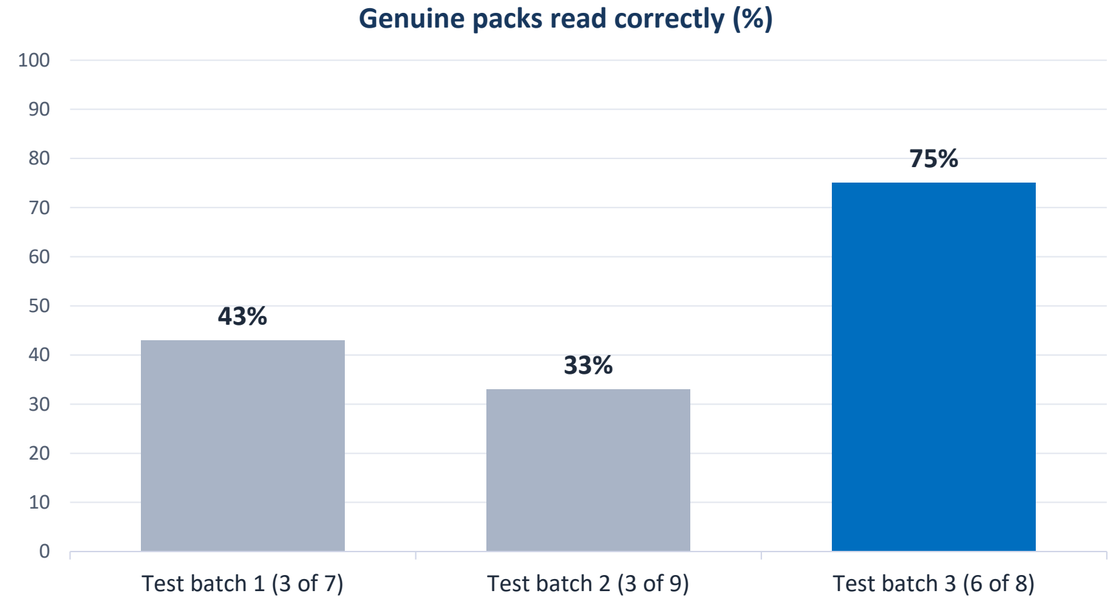
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# FEASIBILITY AND VIABILITY

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## What we've built and tested

| 287 | 75 | 35 | 2 of 2 |
|---|---|---|---|
| backend tests | mobile app tests | real label scans | planted violations caught |




*Every false alarm, we tracked down, fixed and added a test for. 35 scans on 4 real products is a small sample, so we don't call it a benchmark.*

## What could go wrong, and what we do

| Risk | Likelihood | Impact | How we handle it |
|---|---|---|---|
| Photo is blurry, shiny or taken too far away | High | Medium | Anything we can't read goes to REVIEW, not FAIL, so a person checks it |
| OCR misreads text, like “Rs.109,00” | High | Medium | The app handles common slips; the user can correct fields, and every edit is logged |
| A product isn't in our registry yet | High | Low | We show it as “not verified”, and no complaint is raised against the brand |
| Angled photo gives a wrong font size | Medium | Low | If the size reading looks doubtful, it goes to REVIEW instead of FAIL |
| Someone fakes or changes the evidence | Low | High | Each photo is fingerprinted and time-stamped, and a sent scan can't be edited |

## Where we stand

**TRL 4**

tested by our team

- **Working:** the app, server and dashboard all work together
- **Tested:** on 35 real label scans of 4 products
- **Registry:** 8 products, 3 of them from real packs
- **Next:** load Rule 27 registration data, big FMCG brands first

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# IMPACT AND BENEFITS

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## Who benefits

### Consumers

Scan a pack right in the shop. If something's off, the complaint is filed for you, with your photo as proof.

### Inspectors

They only get cases with real evidence, pinned on a map, with the report already filled in.

### Controller of Legal Metrology

The photo, fingerprint check, GPS and time, all on one screen. Then confirm, reject or send the notice.

### Honest manufacturers

When we're unsure, it's REVIEW, not FAIL, and two people must agree, so genuine packs don't get wrong notices.

## How LMCS compares

| | Reads label | LM rule checks | Registry tamper check | Evidence to notice |
|---|---|---|---|---|
| **LMCS** | ✓ | ✓ | ✓ | ✓ |
| **Label checkers** (LabelVeda, Esko Comply) | ✓ | ✓ | – | – |
| **Govt portals** (eMaap, TN LMCTS) | – | – | – | Partly |

*Based on what these tools say about themselves publicly (details in our proposal).*

## Roadmap

PHASE 1 · BUILT

### Core system

- App, server and dashboard
- 13 rules, evidence trail
- Tested on 35 real scans

PHASE 2

### Registry & reach

- Registry from Rule 27 data
- Hindi and regional labels
- Offline mode for inspectors
- Cloud hosting over HTTPS

PHASE 3

### Pilot

- Try it with a Legal Metrology office
- Trained FSSAI-mark detector
- Check e-commerce listings

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# RESEARCH AND REFERENCES

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| Reference | How we used it |
|---|---|
| **Legal Metrology (Packaged Commodities) Rules, 2011** | Our 13 checks cite Rules 6, 9 and 27, and so does the GO notice |
| **Machine vision & product code recognition** pmc.ncbi.nlm.nih.gov/articles/PMC10521108 | How OCR and text detection work on packaging |
| **Packaged Commodities Registration portal** lm.doca.gov.in/pcr/certificates | Where a real product registry would get its data (Rule 27) |
| **Grocery label detection & recognition** mdpi.com/2076-3417/13/5/2871 | What goes wrong with label photos: glare, curves, messy layouts |
| **Google ML Kit: Text Recognition v2 and Barcode Scanning** | Reads the label text (with line positions) and the barcode on the phone |
| **N. Otsu (1979), threshold selection from gray-level histograms** | OpenCV uses Otsu's method to split ink from paper for the Rule 10 contrast check |
| **GS1 General Specifications** | An EAN-13 barcode is 37.29 mm wide, so we use it as a ruler for Rule 9 |
| **WCAG 2.1 contrast ratio** w3.org/TR/WCAG21 | Where our 4.5:1 readability line for Rule 10 comes from |

## Prototype Demo Video Link

**▶ Watch Working Prototype**

## Scan Smart. Verify Right. Protect All